

EmoMirror: Negotiable Affective Mirroring for Students with Elevated Alexithymia

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Abstract

Expressive writing can support mental health reflection, but it assumes that writers can recognize and name what they feel. Students with elevated alexithymia tendencies often face emotion vocabulary poverty: they experience distress yet struggle to distinguish, express, and revisit affective states. We examine how affective journaling systems can support such writers without turning uncertain model inference into an emotional verdict. We conducted a formative study with nine university students with elevated alexithymia tendencies, identifying three requirements: low-friction movement from capture to reflection, AI feedback as negotiable hypotheses, and user control over feedback timing and intensity. We introduce EmoMirror, a journaling prototype that renders model-inferred affect through kinetic typography, candidate emotion labels, valence-arousal fine-tuning, saved records, body-sensation logging, and user-triggered reflection. A within-session evaluation with eight students provides preliminary design insights. Participants used editable labels to move from vague affect toward more specific emotion language, engaged more readily in cause-focused reflection than hidden self-disclosure, and treated affective mismatch as useful only when they could revise, delay, or reject cues. EmoMirror shows how real-time affective typography can operate as negotiable scaffolding for emotional sense-making while preserving user interpretive authority.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Interaction design**; • **Applied computing** → *Psychology*; Health informatics.

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Keywords

alexithymia; affective journaling; affective computing; kinetic typography; emotion reflection; human-AI interaction; mental health

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1 Introduction

Emotion journaling can support reflection only when writers have words for what they feel. Expressive writing and journaling help people organize emotional experiences and make sense of stressful events [11, 12]. Yet these practices assume that writers can recognize, name, and articulate their emotions. This assumption breaks down for students with elevated alexithymia tendencies, who often experience distress as an undifferentiated sense of being unwell. Alexithymia describes difficulties identifying and describing feelings [1]; in everyday journaling, these difficulties appear as emotion vocabulary poverty, uncertainty about bodily and affective cues, and friction when turning felt experience into language. This paper examines how an affective journaling system can support these students without treating them as clinical patients or replacing their own interpretation.

Prior work gives mental health journaling systems several building blocks, but leaves open how affective feedback should enter the act of writing. Mental wellness self-tracking studies show that personal records can help users notice emotional patterns, while also raising concerns about burden, interpretation, and data sensitivity [4]. Recent AI journaling systems use large language models, contextual prompts, and multimodal records to support reflection, including systems such as MindfulDiary, DiaryMate, and MindScape [7, 8, 10]. Separately, kinetic and affective typography research shows that animated or adaptive text can communicate emotional qualities through motion, color, scale, and font form [9, 14, 15].



However, much of this work focuses on post-hoc summaries, conversational guidance, reader perception, or system-generated interpretations. These studies leave open how model-inferred emotion should become a real-time, visible, and revisable cue during first-person emotional writing.

The central design challenge is to make uncertain affective inference useful without making it appear certain. Mental health applications can cause harm when they present uncertain outputs as authoritative judgments [5]. This risk matters for alexithymic users because they may want help naming feelings while still needing authority over their own experience. We therefore frame affective feedback as a negotiable hypothesis: a cue the writer can inspect, question, revise, delay, or ignore. This stance aligns with HCI work on ambiguity as a resource, where open-ended representations can support reflection when users receive scaffolds for interpretation [13].

We first conducted a formative study with nine university students who reported elevated alexithymia tendencies. The study identified three design requirements for affective journaling. First, systems should lower the threshold for moving from immediate expression to later reflection. Second, AI feedback should remain a negotiable hypothesis rather than an emotional verdict. Third, the timing and intensity of affective feedback should remain under user control. These findings informed EmoMirror as a reflective journaling prototype rather than an intervention that claims to improve emotion regulation.

We introduce EmoMirror, a negotiable affective mirror for journaling. EmoMirror lets writers capture typed or spoken entries, renders model-inferred affect through real-time kinetic typography, presents candidate emotion labels, supports optional valence-arousal fine-tuning, saves records, and offers user-triggered reflection. For spoken entries, EmoMirror combines semantic and acoustic affect estimates in valence-arousal space. The system turns affective inference into visible and revisable cues: the writer sees a possible interpretation, compares it with their own experience, and decides whether to keep, revise, hide, or revisit it. EmoMirror therefore treats dynamic typography as an interactional scaffold for emotional sense-making while leaving emotional certainty outside the system's scope.

We evaluated EmoMirror in a short within-session mixed-method prototype study with eight university students who reported TAS-20 scores above 52. We report descriptive statistics and interview findings, not efficacy claims. The preliminary evaluation suggests that EmoMirror helped some participants move from vague affect toward more specific emotion labels, prompted cause-focused reflection more consistently than hidden self-disclosure, and made mismatch useful only when participants could revise or reject the cue. We contribute three claims. First, students with elevated alexithymia tendencies need AI-mediated affective feedback they can negotiate. Second, EmoMirror operationalizes affective inference as visible and revisable scaffolding. Third, interpretive control, feedback pacing, and non-diagnostic language are first-order requirements for real-time affective journaling. We next describe the formative study, system design, prototype evaluation, and discussion.

Table 1: Participants in the formative study. All participants were university students.

Participant	TAS-20	Gender	Age	Major
P01	59	Male	20	Electronic Information
P02	53	Female	20	Artificial Intelligence
P03	68	Female	19	Civil Engineering
P04	63	Female	19	Electrical Engineering
P05	61	Female	20	International Economics and Trade
P06	60	Female	19	Artificial Intelligence
P07	60	Female	21	Design
P08	65	Female	22	Intelligent Vehicles
P09	61	Female	21	Primary Education

2 Formative Study

We conducted a formative study to understand how university students with elevated alexithymia tendencies express and revisit emotions, and how they negotiate AI-mediated affective feedback. The study informed EmoMirror's design requirements; it did not evaluate whether the system improves emotion awareness or regulation.

2.1 Participants and Procedure

We recruited nine university students (ages 19–22; TAS-20-C scores 53–68) through campus channels. We used 52, the lower bound of the legacy 52–60 “possible alexithymia” range, as a recruitment threshold [16]. Although the Chinese TAS-20 is validated in student and clinical samples [1, 17], alexithymia is dimensional rather than a categorical diagnosis [2]. We therefore used scores only for eligibility and sample description, not diagnosis, subgroup comparison, or interpretation of individual experiences. None of these participants joined the subsequent prototype evaluation.

Participants completed approximately 30-minute semi-structured interviews, face-to-face or by video, about emotion expression, difficulties identifying and describing emotions, journaling, a hypothetical AI-assisted journaling tool, and visual-feedback preferences. We framed EmoMirror as a non-diagnostic system offering possible cues through affective typography, emotion prompts, and optional post-writing reflection. A questionnaire provided descriptive evidence about journaling barriers, feedback preferences, privacy concerns, and desired features.

Participants consented to interview recording and research use of study materials and could skip questions, pause, or withdraw without penalty. We anonymized materials before coding and stored recordings and identifying information for one year in a protected, researcher-only database.

Using reflexive thematic analysis [3], we coded the anonymized structured interview records, consolidated codes iteratively, retained contradictory cases, and developed themes around emotion expression, AI feedback, and design requirements. Because the records were structured rather than verbatim transcripts, we limit the analysis to explicitly documented accounts and design implications. The authors translated all quotations from Chinese.

2.2 Findings and Design Implications

2.2.1 Emotional Journaling Needs a Low-Friction Bridge from Expression to Reflection. Participants did not lack ways to express

emotion. Instead, they often relied on socially situated and immediate channels, such as talking with friends, venting, or posting a brief status. As one participant explained, “Chatting with friends feels the most natural” (P08). However, these channels did not always create a private, durable record that participants could revisit. At the same time, journaling was not a frictionless alternative. Participants described lacking time, motivation, or an established practice; P04 stated, “Laziness is the main reason,” and identified help with *starting* a record as most valuable.

Writing also introduced a tension between capture and reflection. Some participants felt that formal writing was too slow or overly organized to preserve fleeting feelings. P08 noted that focusing on making a record coherent could make it feel “less real.” Yet, once emotions were recorded, revisiting them could create useful distance. P05 described rereading a journal entry “from a third-person perspective” as helping them view their feelings more objectively.

These accounts suggest that EmoMirror should not assume that users are already committed journal writers. Instead, it should bridge fleeting expression and later reflection through lightweight entry points, support for informal or multimodal expression, and mechanisms that preserve records for later review.

2.2.2 AI Feedback Should Remain a Negotiable Hypothesis, Not an Emotional Verdict. Participants frequently described a gap between feeling distressed and understanding what that distress meant. P08 recalled feeling low but being unable to determine “what type of emotion” it was. Participants therefore wanted AI feedback to provide more than a positive-versus-negative classification. They requested clues about emotion type, intensity, and possible causes. P02, for example, wanted to know “how distressed” the system inferred they were and “why” it made that inference.

However, participants did not want the system to convert complex experiences into definitive labels. P01 cautioned that emotions are inherently complex and questioned why they should be reduced to a few fixed words. Similarly, participants resisted systems that would immediately replace a vague expression with a supposedly more accurate emotion word. Instead, they wanted feedback to remain open to interpretation. P07 asked for the system to explain “which sentence” led it to infer an emotion.

Thus, the central value of AI feedback lies not in claiming emotional certainty, but in opening an inquiry. EmoMirror should present affective outputs as provisional cues, expose their basis where possible, and allow users to accept, ignore, question, or revise them. This framing preserves users’ authority over their own emotional interpretations while still supporting reflection on feelings that are difficult to articulate.

2.2.3 Affective Feedback Becomes Supportive Only When Users Can Negotiate Its Pace and Intensity. Participants responded to dynamic visual feedback in sharply different ways. Some anticipated that visible affective changes could interrupt writing or intensify negative emotions. P02 explained that real-time feedback might make them stop writing because it could create a sense of “losing control.” Others preferred post-writing feedback because immediate prompts could redirect their thoughts before they had fully expressed themselves.

In contrast, several participants saw visual affective feedback as an opportunity for release and deeper expression. P07 preferred

feedback after a sentence or short passage, while P08 described dynamic expression as resembling a friend who “helps speak out” their emotions. For this participant, visual amplification could make a feeling more manageable rather than more intense.

These contrasting responses indicate that affective visualization cannot function as a uniform intervention. Its usefulness depends on the writer’s momentary readiness for feedback, the feedback’s intensity, and whether the user can control when it appears. EmoMirror should therefore support multiple feedback rhythms, including on-demand, segment-level, post-writing, and real-time modes. It should also provide low-arousal visual feedback, clear pause and disable controls, and supportive language during negative emotional states.

Across these findings, trust emerged as a cross-cutting requirement. Participants could describe AI as a companion or reflective mirror, but they also worried about being monitored or judged. P02 stated that the idea of “an AI knowing all of my thoughts” felt frightening, while P09 described a supportive system as “more like companionship.” EmoMirror should therefore prioritize local and private interaction, non-diagnostic language, user-controlled feedback, and transparent boundaries around emotional inference.

3 EmoMirror

3.1 Overview of EmoMirror

EmoMirror operationalizes the formative findings as a negotiable affective mirror for journaling. Our formative study identified three requirements: writers need a low-friction bridge from immediate expression to later reflection, AI feedback should remain a negotiable hypothesis, and affective feedback should follow the writer’s pace. The Journal implements these requirements as four system stages: capture, analysis and rendering, negotiable revision, and journal storage. These stages support a recurring writer-facing loop of expressing, sensing, visualizing, adjusting, recording, and reviewing. Figure 2 connects this interaction loop to the underlying emotion-analysis pipeline and shows how saved journals support Body Sensation and Emo Echo.

EmoMirror makes the writer’s interpretation the authority for each affective cue. The prototype includes a Journal interface for lightweight capture, kinetic affective typography, affective captions, coordinated emotion labels, an optional valence-arousal fine-tuning surface, saved records, and user-triggered reflection views. These features, shown in Figure 1, form one workflow across capture, feedback, revision, and review. The design therefore treats affective feedback as material for sense-making: the system offers a cue, and the writer decides how to revise, save, ignore, or revisit it.

3.2 Key Features of EmoMirror

3.2.1 Low-Friction Capture Preserves Fleeting Expression for Later Reflection. Journal separates emotional capture from formal diary writing. Participants described formal journaling as valuable after the fact, but difficult to start during fleeting or messy emotional moments. EmoMirror therefore accepts short typed fragments and spoken entries before writers organize an experience into a coherent account. Text-based affect was estimated using DeepSeek V3 from typed text or Whisper-generated transcripts. For spoken entries, Wav2Vec2 also estimates acoustic affect directly from the

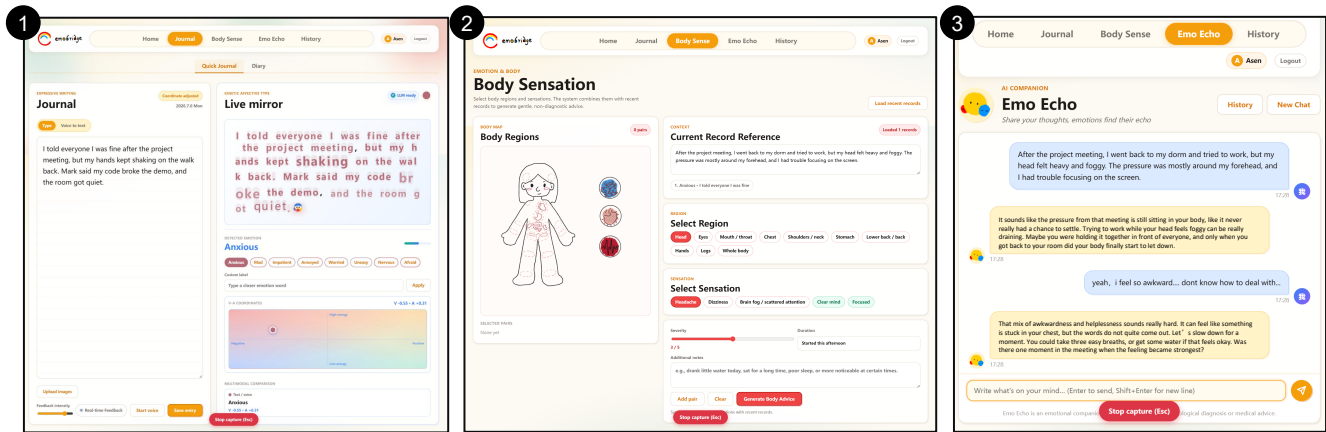
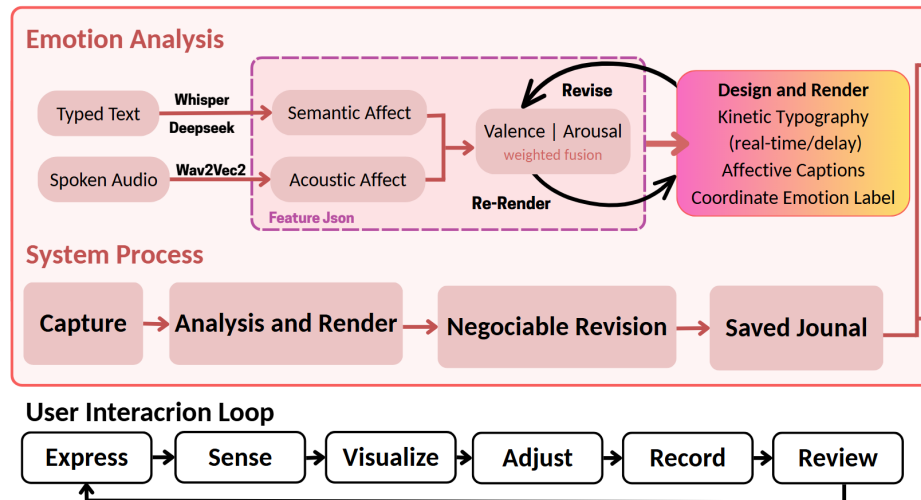
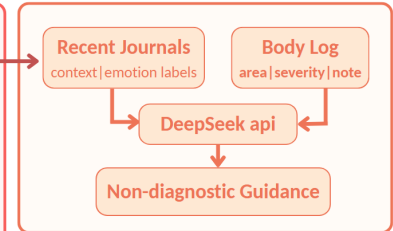


Figure 1: User-facing surfaces of EmoMirror. (1) Journal pairs expressive writing with a live kinetic mirror, candidate labels, and V-A adjustment so writers can inspect and revise model cues during capture. (2) Body Sensation records bodily sensations with recent journal context as optional reflection material. (3) Emo Echo provides user-triggered conversational reflection within a non-diagnostic framing.

A. EmoMirror Journal



B. Body Sensation



C. Emo Echo

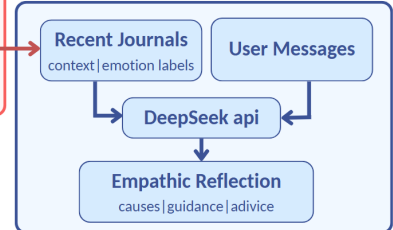


Figure 2: EmoMirror’s interaction loop and processing pipeline. (A) Text or Whisper transcripts feed DeepSeek semantic-affect analysis, while Wav2Vec2 analyzes spoken audio. Weighted valence–arousal fusion drives real-time or delayed kinetic typography, affective captions, and emotion labels that writers can revise before saving and reviewing entries. (B) Body Sensation combines recent journals with a body log for non-diagnostic guidance. (C) Emo Echo combines recent journals and user messages for empathic reflection.

audio. Each branch produces a valence–arousal estimate; when both branches are available, weighted fusion combines them into the representation stored in the feature JSON. Typed-only entries use the semantic estimate without an acoustic contribution.

Each Journal fragment becomes a saved record that writers can revisit later. This design preserves the immediacy of unfinished expression while still creating a durable path toward reflection. Diary functions as a later organizing space, where writers can

return to saved material and request structured prompts about events, feelings, possible needs, and reflection questions. EmoMirror therefore turns journaling into a staged workflow: capture first, organize later.

3.2.2 Kinetic Affective Typography Externalizes Cues While Preserving the Writer’s Words. Kinetic affective typography makes model-inferred affective cues visible while preserving the writer’s original text. The design-and-render stage converts the feature JSON into

kinetic typography, an affective caption, and an emotion label coordinated with the current valence–arousal estimate. Writers can receive this feedback in real time or after a delay. Both timing modes render provisional cues through motion, scale, color, weight, and local emphasis. Motion and scale express energy; color and weight suggest affective valence or tension. The text remains the writer’s own sentence, while the rendering creates a visible cue that the writer can inspect.

This feature responds to participants who wanted more than positive–negative sentiment but resisted definitive emotion labels. EmoMirror uses typography to make a possible interpretation perceptible before turning it into a fixed word. Prior work shows that kinetic typography and adaptive font rendering can communicate affective qualities through animated text [9, 14, 15]. EmoMirror redirects that design space toward first-person reflection by asking the writer to compare the rendered cue with their own account.

3.2.3 Candidate Labels and Optional V-A Fine-Tuning Keep Interpretation Negotiable. Candidate labels turn affect labeling into a revision process. After generating a provisional cue, EmoMirror surfaces a primary emotion label and nearby candidate labels. These candidates give writers a small vocabulary for moving between related feelings, such as low-energy negative states and high-energy negative states. Writers can select a candidate label, keep the current label, or revise the cue through the optional valence–arousal surface. Each adjustment returns the revised interpretation to the design-and-render stage, which re-renders the journal entry before the writer saves it.

The V-A surface serves as an optional fine-tuning control that remains secondary to journaling. The interface presents the horizontal axis as negative-to-positive and the vertical axis as low-to-high energy, avoiding technical psychological terminology in the writer-facing view. Dragging the point updates the selected label and candidate labels; selecting a label updates the point. The saved entry records the writer’s negotiated interpretation as the authoritative account for that entry.

This bidirectional revision loop preserves interpretive authority. Participants wanted AI feedback to open inquiry while leaving room to disagree, especially when emotions felt mixed or difficult to name. Candidate labels provide natural-language scaffolding, while V-A fine-tuning supports writers who want more control over nuance. This stance aligns with HCI work that treats ambiguity as productive when users receive scaffolds for interpretation [13].

3.2.4 User-Triggered Reflection Lets Writers Revisit Cues Across Time. EmoMirror distributes reflection across immediate, entry-level, and later review. During writing, Journal offers kinetic typography, affective captions, emotion labels, and optional V-A fine-tuning. Once a writer saves an entry, the system makes its recent context and negotiated emotion labels available to Body Sensation and Emo Echo.

Body Sensation combines recent journal context with a self-reported body log containing the affected area, severity, and an optional note. A user-triggered DeepSeek request turns these inputs into non-diagnostic guidance. Emo Echo combines recent journal context and emotion labels with the writer’s current message. Its user-triggered DeepSeek request returns an empathic reflection organized around possible causes, guidance, and actionable advice.

This temporal structure responds to participants’ conflicting preferences about feedback timing. Some participants expected immediate visual feedback to support expression; others worried that it would interrupt writing or amplify negative affect. EmoMirror therefore offers real-time or delayed rendering and keeps deeper reflection user-triggered. The interaction remains a writer-paced loop from expression and adjustment to recording, review, and renewed expression.

3.3 Safety and Data Boundaries Keep the Mirror from Becoming a Judge

EmoMirror constrains affective mirroring through non-diagnostic language, user-triggered reflection, and scoped access to personal records. The interface presents affective outputs as possible cues for writer interpretation. When users record physical discomfort in Body Sensation or discuss negative moods in Emo Echo, the system responds with empathetic language and brief self-care suggestions, such as paced breathing or taking a warm shower. We designed these responses as supportive reflection, not as professional-help triage or crisis intervention.

EmoMirror also limits how personal records become visible across roles. Ordinary views return records associated with the current participant code, while research-facing access requires separate permission. Deeper AI-generated reflection appears only after the writer requests it. These boundaries answer the formative study’s trust concern: affective journaling becomes threatening when AI appears to monitor, judge, or overclaim. EmoMirror instead positions the system as a revisable mirror whose authority remains subordinate to the writer’s interpretation, responding to documented risks in mental health applications that present uncertain outputs as authoritative judgments [5].

4 User Study and Results

4.1 Study Design

We conducted a short, within-session, mixed-method evaluation with eight university students (ages 19–20; TAS-20-C scores 53–80, $M = 62.50$, median = 60.50) who had not joined the formative study. The study examined perceived support for emotion naming, reflective writing, and negotiation of model-generated cues. Participants completed the validated Chinese TAS-20 [1, 17]; we used scores of 52 or higher for eligibility only [16] and treated them as continuous and non-diagnostic, without clinical grouping [6].

Participants first received a short introduction to EmoMirror and then wrote about emotionally meaningful experiences. They observed kinetic affective typography and provisional emotion labels, revised cues through candidate labels or the valence–arousal control, and compared the experience with ordinary memo-style journaling. After the session, participants completed an 11-item, 1–5 Likert questionnaire, a separate feedback-timing item (T1), and a brief semi-structured interview. Table 2 triangulates the principal questionnaire patterns with interview evidence and design implications. Appendix A.2 provides the author-translated items, and Appendix A.3 reports the complete item-level distributions. The item groups organize the analysis and do not represent validated

Table 2: Mixed-method synthesis of the preliminary prototype evaluation ($N = 8$).

Result pattern	Questionnaire evidence	Interview evidence	Design implication
Editable cues supported emotion naming unevenly	Q3 drew 5/8 agreement for differentiating vague or mixed feelings. Q2 split evenly: four participants selected 1–2 and four selected 4–5.	Participants valued candidate labels when they moved beyond vague feelings, but several described the label set as broad, sparse, or insufficient for mixed emotions.	Treat affective cues as an editable label space through candidate labels, multi-label selection, V-A adjustment, and segment-level revision.
Cause-focused reflection exceeded deeper disclosure	Q5 drew 5/8 agreement for identifying causes or triggers, whereas Q6 drew 2/8 for salient or unspoken content.	Some participants continued writing to analyze causes; others described initial engagement without deeper disclosure or identified privacy as a boundary.	Use optional reflection prompts and credible privacy controls; preserve user pacing and avoid pressuring users to disclose sensitive content.
Interpretive control made mismatch negotiable	Q8 drew 4/8 agreement for productive mismatch, while Q9 drew 3/8 for misreading or resistance. Q10 and Q11 each drew 7/8 agreement. For T1, 6/8 preferred real-time feedback and 2/8 preferred feedback after the full entry.	Mismatch sometimes prompted self-inquiry, but premature or salient feedback could feel like misreading or interruption. Participants requested control over labels, timing, and presentation intensity.	Provide revision, timing, intensity, hiding, and explanation controls so users retain interpretive authority over model-generated affective cues.

psychometric factors. We coded interviews with a lightweight deductive codebook aligned with the three evaluation questions while retaining negative cases. We translated all quotes from Chinese.

Before participation, all participants signed informed-consent forms covering interview recording, scientific use of journal materials, and transmission of journal content to external model services, including DeepSeek. Participants could skip tasks or questions, pause, or withdraw without penalty. We anonymized all study materials and stored recordings, journal entries, and identifying information for one year in a protected database accessible only to the study researchers.

EmoMirror provided supportive, non-diagnostic responses when participants recorded physical discomfort or discussed negative moods in Emo Echo. These responses used empathetic language and brief self-care suggestions, such as paced breathing or taking a warm shower. We did not log these responses as separate events or implement a researcher-led distress-escalation protocol. We therefore describe this mechanism as supportive feedback, not a validated crisis safeguard, and cannot quantify how often it appeared during the study.

Table 2 reports selected mixed-method contrasts; agreement denotes ratings of 4 or 5, and Q9 retains its original negative direction. Appendix A.3 preserves the complete response distributions.

4.2 Evaluation Findings

4.2.1 Editable Cues Supported Emotion Naming. Emotion-naming support varied across participants (Table 2). P04 valued receiving “more specific judgments and cues” when an emotion initially felt vague, and P07 used nearby candidates when the first label did not fit. P05 instead wanted finer-grained, multi-label feedback because emotions could be mixed. **Takeaway.** The interaction around a label mattered more than any single inference; candidate exploration and revision should remain first-order controls.

4.2.2 Cause-Focused Reflection Exceeded Disclosure. EmoMirror prompted cause-focused reflection more consistently than deeper disclosure (Table 2). P04 explained, “I would keep writing to analyze why I had this emotion.” P08 found the interaction engaging but did not disclose content they would otherwise omit, while P05 would not fully open up on an unfamiliar platform. **Takeaway.** Reflection

prompts should remain optional and operate within credible privacy boundaries and user-controlled pacing.

4.2.3 Interpretive Control Made Mismatch Negotiable. Interpretive agency produced the clearest pattern (Table 2). P08 used disagreement as a prompt to “search for my real feeling,” but P05 found premature labels “too fast” and “too arbitrary.” P07 also described salient real-time feedback as visually noisy. **Takeaway.** Mismatch became reflective only when users could reject, revise, delay, or soften it. The study therefore supports a bounded design claim: affective journaling systems should present model outputs as revisable cues that preserve the writer’s interpretive authority.

5 Discussion

This paper argues that affective journaling for students with elevated alexithymia should not make emotion recognition more authoritative; it should make uncertain affective cues easier to inspect, revise, and revisit. The formative study identified three requirements: low-friction capture, AI feedback as a negotiable hypothesis, and user control over timing and intensity. EmoMirror operationalizes them through lightweight text/speech capture, kinetic affective typography, candidate labels, optional V-A tuning, saved records, and user-triggered reflection. The prototype evaluation provides preliminary support: participants valued EmoMirror when it helped them move from vague affect toward specific language, reflect on causes, and disagree with mismatched cues without surrendering interpretive authority.

The workshop-level contribution is therefore a bounded design claim: negotiable affective mirroring can turn model-inferred emotion into reflective material, not emotional verdicts. For mental health journaling, this distinction matters because deeper disclosure, diagnostic language, or always-on feedback can threaten privacy and trust. EmoMirror suggests that interpretive control, pacing, revision, and non-diagnostic boundaries are core design requirements.

Limitations include eight participants, a short within-session evaluation, descriptive statistics, and perceived rather than longitudinal outcomes. Future work should examine repeated use, feedback timing, multi-label and segment-level revision, and changes in emotion awareness over time.

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A Appendix

A.1 Toronto Alexithymia Scale (TAS-20)

A.1.1 Chinese-Language Version and Study Role. The 20-item Toronto Alexithymia Scale (TAS-20) measures three self-reported facets of alexithymia: difficulty identifying feelings (DIF), difficulty describing feelings (DDF), and externally oriented thinking (EOT) [1]. Participants in both studies completed the validated Chinese-language version (TAS-20-C). Zhu et al. developed this version through translation, back-translation, and psychometric evaluation in Chinese student and clinical samples [17].

We confined the TAS-20 to participant recruitment and sample description. The study-specific questionnaire and interviews assessed participants’ experiences with EmoMirror. This appendix therefore documents the TAS-20 version, scoring procedure, and recruitment threshold. The original instrument remains the authoritative source for individual item wording [1].

A.1.2 Response Format and Scoring. The TAS-20 contains 20 statements rated on a five-point response scale.

Score	Response anchor
1	Strongly disagree
2	Disagree
3	Neither disagree nor agree
4	Agree
5	Strongly agree

We reverse-scored Items 4, 5, 10, 18, and 19, then summed all 20 responses. Total scores range from 20 to 100; higher totals indicate higher levels of self-reported alexithymia tendencies [1]. Table A1 reports the factor composition used for scoring.

Table A1: TAS-20 factor composition and score ranges.

Factor	Construct	Items	Range
DIF	Difficulty identifying feelings	1, 3, 6, 7, 9, 13, 14	7–35
DDF	Difficulty describing feelings	2, 4, 11, 12, 17	5–25
EOT	Externally oriented thinking	5, 8, 10, 15, 16, 18, 19, 20	8–40

A.1.3 Recruitment Threshold and Non-Diagnostic Interpretation. We set 52 as the recruitment threshold because it marks the lower boundary of the conventional 52–60 “possible alexithymia” band [16]. This criterion identified students with elevated self-reported tendencies while retaining the intermediate score range. Table A2 presents the historical interpretive bands for transparency.

Table A2: Historical TAS-20 interpretive bands.

Total score	Historical interpretation
20–51	Lower or no alexithymia tendencies
52–60	Possible alexithymia tendencies
61–100	Higher alexithymia tendencies

The scale developers now characterize alexithymia as a continuous personality trait and caution against treating cutoff groups as

qualitatively distinct categories [2]. We therefore used scores of 52 or higher only for recruitment. We interpret all observed TAS-20 scores descriptively and non-diagnostically.

A.2 EmoMirror Post-Study Questionnaire

A.2.1 Administration and Analytical Role. Participants completed a study-specific questionnaire in Chinese after using EmoMirror. Q1–Q11 used a five-point scale from 1 (strongly disagree) to 5 (strongly agree). T1 separately recorded feedback-timing preferences. The English wording below represents an author translation of the administered Chinese items.

The questionnaire examined three parts of the user experience: emotion naming, reflective writing and self-disclosure, and negotiation of affective feedback. These theory-driven groups organize reporting; they do not constitute validated subscales. We report raw item distributions and the number of participants selecting 4 or 5; T1 remains a separate categorical preference item.

A.2.2 Emotion Naming Support (Q1–Q3). Q1–Q3 examined whether EmoMirror helped participants notice, name, and differentiate emotions more specifically.

Q1 — Noticing and describing emotions. Compared with journaling in an ordinary memo app, EmoMirror helped me notice which emotions I might have been experiencing and describe and understand them more specifically.

Q2 — Expanding emotion vocabulary. The emotion labels provided by EmoMirror helped me think of emotion words that I had not originally written and further notice and understand my emotional state.

Q3 — Differentiating vague or mixed feelings. Emotion-label recognition, candidate-label revision, and adjustments on the emotion coordinate panel helped me distinguish vague or easily confused feelings, such as feeling bad, annoyed, or okay.

A.2.3 Reflective Writing and Self-Disclosure (Q4–Q7). Q4–Q7 examined continued writing, reflection on emotional causes, sensitive self-disclosure, and perceived expressive fit.

Q4 — Continued writing. Compared with ordinary memo journaling, EmoMirror made me more willing to continue writing instead of simply recording what happened.

Q5 — Cause-focused reflection. Compared with ordinary memo journaling, EmoMirror helped me further recognize the causes or triggers underlying my emotions.

Q6 — Sensitive self-disclosure. Compared with ordinary memo journaling, EmoMirror made it easier to write about what truly mattered to me, such as needs, grievances, fears, expectations, or unspoken thoughts.

Q7 — Perceived expressive fit. After using EmoMirror, my entry felt closer to the feelings I truly wanted to express than an entry in an ordinary memo app.

A.2.4 Affective Feedback and Interpretive Control (Q8–Q11). Q8–Q11 examined whether mismatched affective cues prompted reflection, produced a sense of being misread, or increased the need for revision and interpretive control.

Q8 — Productive mismatch. When the emotion inferred by the system did not fully match my own feeling, the discrepancy sometimes helped me reinterpret and further understand myself.

Q9 — Misreading and resistance risk. When the emotion inferred by the system did not match my own feeling, I felt that the system was misreading me, which created cognitive dissonance and resistance.

Q10 — Revision and interpretive authority. I considered it important to revise the system-generated emotion label or emotion coordinates; I would not want to accept the system’s judgment without being able to change it.

Q11 — Cue instead of verdict. Overall, EmoMirror felt more like a reference cue about my emotions than a system that judged my true emotion for me.

Q9 captures perceived misreading or resistance. We report it in its original direction and keep it separate from supportive-feedback summaries; higher responses therefore indicate a stronger negative experience.

A.2.5 Feedback Timing and Background Fields. T1 asked, “Which timing for kinetic-text feedback and emotion labels felt most appropriate and comfortable?” Response options were: shown in real time; shown after completing a paragraph; and shown after completing the full journal entry.

The questionnaire also recorded gender, occupation, academic major, age, and TAS-20 total score. We used administrative identifiers only to match questionnaires with interviews and excluded them from the research appendix and published results.

A.2.6 Scoring and Reporting Boundaries. Table A3 summarizes the analytical groupings used to organize the user-study results. T1 and the background fields remain outside these groupings.

Table A3: Study-specific questionnaire groupings.

Analytical group	Items	Reporting approach
Emotion Naming	Q1–Q3	Item distributions and agreement counts
Reflection and Disclosure	Q4–Q7	Item distributions and agreement counts
Mismatch and Interpretive Agency	Q8–Q11	Item distributions and agreement counts; Q9 retains its original negative direction
Feedback Timing	T1	Frequency and percentage by response option

The questionnaire captures participants’ perceived short-term experience with the prototype. Its item-level responses support descriptive design findings about emotion naming, deeper writing, mismatch, and interpretive control. Figure A1 reports the complete distributions underlying the selected contrasts in Table 2. Clinical efficacy, long-term emotion regulation, and the accuracy of inferred emotions remain outside the study’s measurement scope.

A.3 Item-Level Questionnaire Distributions

Figure A1 preserves all five response categories for Q1–Q11 and complements the condensed evidence synthesis in Table 2. The three panels organize items by analytical role rather than validated

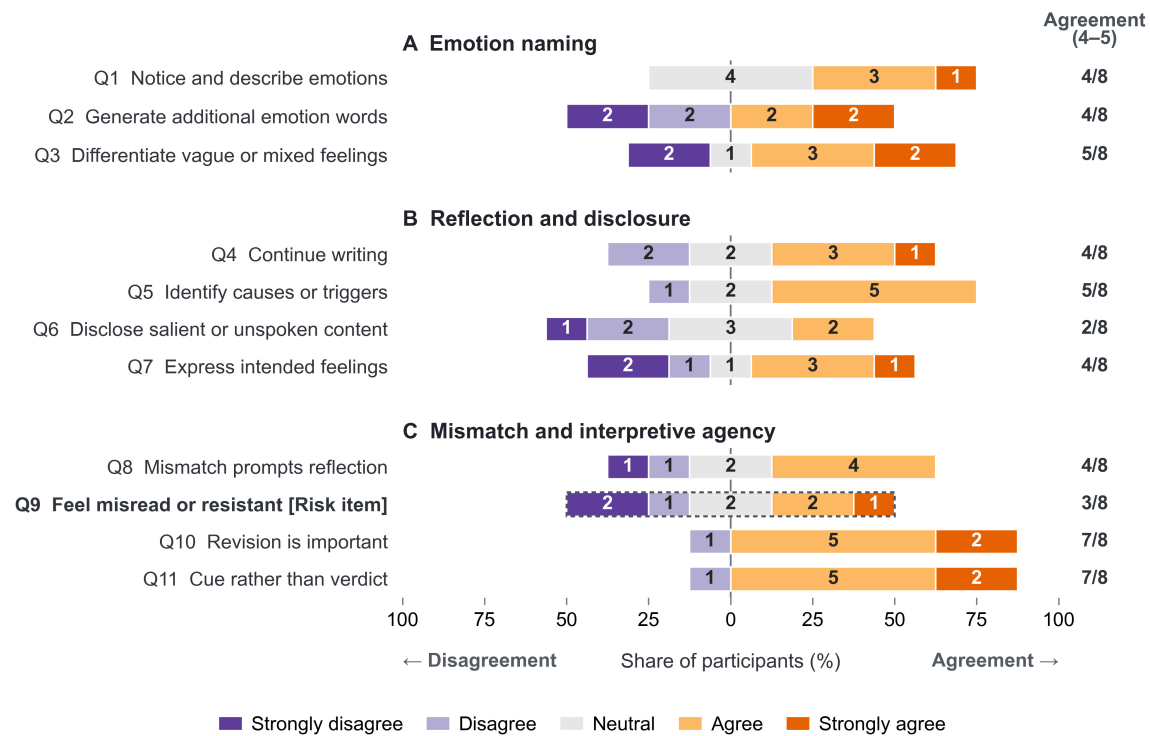


Figure A1: Item-level responses to the post-session questionnaire ($N = 8$). Diverging bars show raw ratings on five-point Likert scales, centered on the neutral category. In-segment labels give participant counts, and the right-hand column reports agreement (ratings 4–5). The plot retains Q9 in its original negative direction, so agreement denotes perceived misreading or resistance.

factors. Q9 remains in its original negative direction, and T1 remains outside the plot because it records a categorical timing preference.

The distributions clarify the heterogeneity summarized in Table 2. Q10 and Q11 show the most concentrated agreement, with seven of eight participants selecting 4 or 5. Q2 divides evenly between disagreement and agreement, while Q5 exceeds Q6, separating cause-focused reflection from deeper disclosure. Q9 also shows

that three participants experienced misreading or resistance. These descriptive patterns motivate the design implications in Table 2 but do not establish psychometric factors or prevalence beyond this sample.